Software Requirements Specification (SRS)

AI-Powered Predictive Maintenance System

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1. Introduction & Glossary

1.1 Purpose

This document defines the comprehensive software requirements for the AI-Powered Predictive Maintenance System, specifying functional capabilities, user interactions, quality attributes, and operational constraints.

1.2 Scope

The system processes time-series equipment sensor streams (vibration, temperature, pressure) to detect anomalies, forecast Remaining Useful Life (RUL) using machine learning and deep learning models, provide explainable AI insights (SHAP/LIME), power

an interactive Streamlit digital twin dashboard, execute an LLM-based maintenance assistant, and generate automated diagnostic reports.

1.3 Glossary & Acronyms

Term / Acronym	Definition
RUL	Remaining Useful Life
XAI	Explainable Artificial Intelligence (SHAP / LIME)
FFT	Fast Fourier Transform
API	Application Programming Interface
SRS	Software Requirements Specification
DFD / ERD	Data Flow Diagram & Entity Relationship Diagram

2. Overall Description & Stakeholders

2.1 Product Perspective & Assumptions

- The system is deployed in a mid-sized manufacturing plant operating rotating equipment (pumps, motors, and compressors).
- Sensor data (vibration, temperature, pressure) is provided via a simulated historical/public data stream. There is no direct live SCADA connection in this phase.
- System Constraint: The system operates strictly as a decision-support system. It never controls machines directly or writes back commands to operational control/PLC systems.

2.2 Stakeholder Register Summary

- Primary Users: Maintenance Technicians (SH-01), Reliability Engineers (SH-02), Machine Operators/Shift Supervisors (SH-03).
- Secondary / Technical Users: Maintenance Planners (SH-04), Plant Managers (SH-05), Finance (SH-06), HSE Officers (SH-07), ML Engineers (SH-08), IT Administrators (SH-09), OT Engineers (SH-10).
- External & Academic: Equipment OEMs, Cloud/LLM Providers, DEPI Evaluators & Project Team (SH-11 to SH-16).

3. System Functional Requirements (FRs)

3.1 AI/ML & Core Predictive Analytics Requirements

- FR-AI-01 — Sensor Data Validation: Validate incoming fields, types, missing values, timestamps, and operating ranges before processing.
- FR-AI-02 — Time-Series Data Preprocessing: Handle missing data, noise reduction, normalization, windowing, and Fast Fourier Transform (FFT) for vibration signals.
- FR-AI-03 — Feature Engineering and Extraction: Generate statistical and frequency-domain rolling features to support prediction models.
- FR-AI-04 — Anomaly Detection: Identify abnormal equipment behavior using trained models to generate anomaly scores and statuses.
- FR-AI-05 — Remaining Useful Life (RUL) Prediction: Estimate remaining equipment lifespan using validated time-series models (LSTM, GRU, 1D-CNN).

- FR-AI-06 — RUL Uncertainty Estimation: Provide a confidence/uncertainty range alongside each RUL prediction (e.g., 18 days [14–23 days]).
- FR-AI-07 — Machine Health Severity Classification: Classify machine health into standard severity levels: Normal, Warning, Critical.
- FR-AI-08 — Explainable AI (XAI): Integrate SHAP and LIME modules to output main contributing sensor factors for warning/critical predictions.
- FR-AI-09 — Configurable Thresholds: Allow authorized users to adjust anomaly, RUL, and sensor limit thresholds without retraining models.
- FR-AI-10 — AI/ML Model Inference: Execute model pipelines via FastAPI backend endpoints, returning structured prediction payloads.
- FR-AI-11 — Model Performance Evaluation: Evaluate candidate models using Precision, Recall, F1, ROC-AUC, MAE, and RMSE metrics.
- FR-AI-12 — Model Comparison and Selection: Select optimal models based on evaluation logs, reliability, and explainability.
- FR-AI-13 — Model Versioning & Tracking: Track experiments, parameters, datasets, and versions using MLflow.
- FR-AI-14 — Data Drift Monitoring: Monitor incoming data distribution shifts against training data to flag review alerts.
- FR-AI-15 — Model Retraining: Support model retraining upon drift detection or performance degradation while retaining older versions.

- FR-AI-16 — Prediction Logging: Record prediction instances, timestamps, model versions, and outputs for auditing.

3.2 UI, Digital Twin, LLM & Reporting Requirements

- FR-UI-01 — Role-Based Dashboard: Provide a Streamlit interface adapting views and fleet overviews based on user roles.
- FR-UI-02 — Machine Detail & Time-Series Charts: Plot historical and real-time sensor streams and RUL trends interactively.
- FR-UI-03 — XAI Visualization: Render SHAP/LIME feature importance graphs beside predictions.
- FR-UI-04 — Digital Twin / What-If Simulator: Enable simulation sliders to test parameter changes and observe RUL updates safely in a sandbox mode.
- FR-UI-05 — Alert Management Table: Display filterable alerts allowing users to *Acknowledge* or *Escalate* statuses.
- FR-LLM-01 — Conversational Maintenance Assistant: Provide a chat interface backed by LangChain for natural language querying.
- FR-LLM-02 — Context-Aware Grounding (RAG): Automatically inject the selected machine's live RUL, score, and logs into LLM prompts to prevent hallucinations.
- FR-LLM-03 — Assistant Guardrails: Enforce strict system prompts prohibiting unsafe commands, out-of-scope advice, or equipment control.

- FR-LLM-04 — LLM Fallback Mechanism: Support automatic fallback from cloud APIs (e.g., OpenAI) to local models (Hugging Face) for privacy or uptime.
- FR-REP-01 & 02 — Automated PDF Diagnostic Reports: Compile health summaries, XAI charts, and LLM-generated narratives into downloadable PDF reports.
- FR-REP-03 — ROI & Cost Calculator: Provide a calculator tab computing downtime cost savings achieved via predictive maintenance.

4. System Behavior & User Stories (Summary)

- Authenticated Access (US-01, UC-01): Role-based login ensuring secure entry and restricted page access.
- Monitoring & Alerts (US-02 to 10, UC-02 to 09): Real-time sensor ingestion, inference processing, severity evaluation, and technician acknowledgement workflows.
- Analysis & Simulation (US-11 to 14, UC-10 to 12): Reliability engineering workflows for RUL range analysis, XAI review, What-if testing, and threshold configuration.
- Reporting & Auditing (US-15 to 24, UC-13 to 16): Plant manager ROI reviews, monthly scheduling, HSE alert history audits, and false-alarm flagging.
- ML Operations (US-25 to 29, UC-17 to 19): Model training, metric comparison, MLflow tracking, drift checking, and version deployment.
- Administration (US-30, 31, UC-20, 21): User management, role assignments, and encrypted API key/local mode configurations.

5. Non-Functional Requirements (NFRs) & Constraints

5.1 Performance:

- Model inference responses within 3 seconds; dashboard fleet overview loads within 5 seconds; standard PDF generation within 10 seconds.

5.2 Accuracy & Reliability:

Deployed models must satisfy evaluation thresholds; uncertainty/confidence ranges must accompany RUL forecasts; invalid sensor records are filtered out.

5.3 Explainability & Usability:

XAI explanations provided for all critical/warning states; role-based tablet-friendly UI with intuitive Normal/Warning/Critical color coding.

5.4 Security & Data Privacy:

Role-based access control (RBAC); secure, non-plain-text API key storage; local-only LLM mode option to prevent sensitive plant data exposure.

5.5 Availability, Maintainability & Scalability:

High operational availability with local fallback options; model versioning and experiment traceability via MLflow; modular architecture supporting additional machine streams.

5.6 Auditability & Cost Control:

Comprehensive logging of prediction events, user actions, and threshold updates; monitoring and optimization of external API usage.

5.7 Operational Safety Constraints:

- NFR-SAFE-01: The system shall operate strictly as a decision-support tool and shall never execute direct control actions on physical machinery or control infrastructure.
- NFR-SAFE-02: The LLM assistant and automated modules shall reject any instruction attempting to bypass safety protocols or alter physical control parameters.

6. Requirements Traceability Matrix (RTM)

Stakeholder Need (SH-ID)	User Story (US-ID)	Functional Requirement (FR-ID)	Use Case (UC-ID)
SH-01 (Technician)	US-02, US-03	FR-UI-03, FR-UI-05, FR-AI-07	UC-04, UC-08
SH-02 (Reliability Eng.)	US-11, US-12, US-13	FR-AI-05, FR-UI-02, FR-UI-04	UC-03, UC-11
SH-03 (Operator)	US-07, US-08, US-09	FR-UI-01, FR-AI-04, FR-UI-05	UC-02, UC-08

Stakeholder Need (SH-ID)	User Story (US-ID)	Functional Requirement (FR-ID)	Use Case (UC-ID)
SH-05 (Plant Manager)	US-19, US-20	FR-UI-01, FR-REP-03	UC-02, UC-14
SH-07 (HSE Officer)	US-22, US-23	FR-UI-05 (partial), FR-AI-07	UC-07, UC-15
SH-08 (ML Engineer)	US-25, US-26, US-28	FR-AI-11, FR-AI-13, FR-AI-14	UC-17, UC-18, UC-19
SH-09 (IT Admin)	US-01, US-30, US-31	FR-UI-01, FR-LLM-04	UC-01, UC-20, UC-21
SH-10 (OT Engineer)	US-17	FR-AI-01, FR-AI-02, FR-AI-03	UC-05